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1/14/2025

Team Choice (for now):
- Identifying Connections (2 credits)

To-do: fill out team file for your team by next Tuesday 12 P.M.
- Meet with team

1/28/2025

Previous Semester Algorithm Revision

- Previous algorithm:
  - Only matched students into pairs
  - took into account:
    - Study type
    - Credit Hours
    - major

    Prioritized pairs with all in common, then 2, then 1, then random.

Revision Ideas:

- Introduce weights to each factor
  - assign them based on individual survey results
- Incorporate matching a group of students functionality to our algorithm (not just pairings)

Goals:

- Start revising algorithm, put on Github.
- Can have multiple algorithms to test out which one is better
  - Use A.I. to determine performance

2/3/2025

First Revision of Algorithm
- took into account students' study type
(night owl / earlyriser)
Major,
availability (weekdays/weekends, time of the day)
learner type (Choose from 4)
Study intensity level
Working style (long/short)] similar
preferred environment
Priority → used to assign weights of
importance for each characteristic

Thinking of limiting characteristics down...
① Study type
② availability
③ learner type } Combine
④ working style
⑤ Major / Course Overlaps
⑥ Location (In-person / online)

* for group studying → limit 4 / group

* revised algorithm sent in slack channel,
need to push it to Github later.

* Changes from Maria's algorithm

- Calculates Compatibility score b/t students based on how many traits they have in common
- is better than previous approach b/c previously we were just manually checking between 2 students whether they are 3 / 2 / 1 traits in common

How is ~~So~~ the score calculated?
- user defined weights (from survey)
- for every trait 2 students have in common, add the associating weights of that trait to score

- finally, normalize it so it's out of 100
- for Group ~~Matching~~ Matching:
Choose the top [Group Size] students with the highest compatibility score.

2/11/2025

Demo Presentation

TO-do:

- make a pull request to Bryan's repo
- revise user.txt to include individual users' weights
- Submit Notebook

2/18/2025

Format of new users.txt:

person = {
[name],
[StudyType], ← nightowl, ..
[credit Hours],
[major],
[availability],
[learner type],
[intensity], ← rate 1-5
[priority],
[working style], ← short / long
[environment], ← Home/library/cafe, etc
[gender],
[minor],
[year],
[interests],
} [weights] ← a list of 9 floats

these are not used in our group's algo, but used for facilitating interactions group

indicating "how important" the 9 features are to each student.

* Sum up to 1.

2/25/2025

Weights Logic Brainstorming

Ideas:

high weights

- If 2 people <u>value</u> a feature a lot, then how similar their response to that feature's question should directly contribute to how compatible they are

Alice

Weight matrix: [
[0.3, 0.2, 0.2, 0.1, 0.1, 0.1, 0, 0, 0],
[0.5, 0.2, 0.2, 0.1, 0, 0, 0, 0, 0]

Bob

* Both Alice and B value feature 1 (let's say it's study type) a lot, so find a way to calculate the similarity score between these 2's response and add it to their aggregate compatibility score

repeat for all 9 features

3/4/2025

Ideas for Similarity Score Calculation:

- First, determine the format / range of student responses to each feature:

1. Study Type → String response
2. Credit Hours → numeric
3. Major → String
4. availability → String
5. learner type → String
6. intensity → numeric
7. priority → String
8. Working style → String
9. Environment → String

for numeric, Calculate Similarity like:

$difference = abs(u1 - u2)$

$Similarity = 1 - (difference / max(u1, u2))$

- For string responses, if response is identical, Score = 1, else 0.

If we combine the similarity score for all features for a user, then that is the <u>Unweighted Compatibility Score</u>.

~~Combinating~~ Combining Weights Approach with Similarity Score.

- To account for individual weights chosen by 2 students, I can average them out and use that as the new weight for the similarity score.

Example: Alice has weight 0.5 for credit hours.
Bob has weight 0.1 for credit hours.
Alice takes 20 credits, Bob 14.
Averaged weight $= (0.5 + 0.1)/2 = 0.3$

Similarity Score (for Credit hours) =

$$0.3\left(1 - [(20-14)/20]\right) = 0.21$$

We can follow this algorithm to obtain the similarity score for all 9 feature.

To do:

- revise Similarity Score Calculation in algorithm.py using new formula.

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Testing Goals:

1) Test on Students with Similar features (no weights applied) ☑

2) Test on groups of various Sizes ☑

3) Scalability Testing (aim for 100 student profiles) ☑

4) Check Student testing is still accurate with weights added ☑

Future Goals:

- Compare results between 3 Scenarios:

① algo with no weights
② algo with predetermind weights
③ algo with Student chosen weights (via Survey result)

For case 2, we can use AI to find the weights that can maximize best student pairing.

4/3/2025

Presentation Slides.

- Focus on algorithm weights logic /explanation

Todo:

- Coordinate with teammates
on presentation